

# Ransomware Sentinel

## User Guide

Version 1.3 (development build, feature/automatic-protection)

For Windows 10 and 11 · runs from the Python source distribution

Screenshots are of the real application. Paths shown are from a test profile.

# Contents

|                                      |    |
|--------------------------------------|----|
| Contents.....                        | 2  |
| 1. What Ransomware Sentinel is ..... | 3  |
| The workflow .....                   | 3  |
| 2. First launch .....                | 3  |
| Step 1 — Welcome.....                | 3  |
| Step 2 — Protected locations .....   | 4  |
| Step 3 — Initial scan .....          | 5  |
| Step 4 — Protection active.....      | 5  |
| The guided tour.....                 | 6  |
| 3. Protected Locations .....         | 6  |
| 4. Initial scan.....                 | 7  |
| 5. Protection active .....           | 7  |
| 6. Dashboard .....                   | 8  |
| 7. Alerts.....                       | 9  |
| 8. Incidents.....                    | 10 |
| 9. Investigation .....               | 11 |
| 10. Quarantine .....                 | 12 |
| 11. Recovery readiness .....         | 13 |
| 12. Diagnostics and audit.....       | 14 |
| 13. Safe Demo (Demo Center).....     | 15 |
| 14. Settings.....                    | 16 |
| 15. Troubleshooting .....            | 17 |
| 16. Limitations.....                 | 17 |

(If the contents list is empty, right-click it in Word and choose Update Field.)

# 1. What Ransomware Sentinel is

Ransomware Sentinel watches the folders where your own files live and recognises the behaviour typical of ransomware: bursts of file changes, mass renaming, and renames to encrypted-looking extensions such as .locked. It gives that activity a risk score from 0 to 100, explains every point of the score, groups related alerts into one incident, and gives you tools to investigate, respond and keep a tamper-evident record of what was done.

Everything runs on your computer. There is no account, no cloud service and no telemetry upload.

**What it is not.** Sentinel is not antivirus and does not replace Microsoft Defender. It has no kernel-level monitoring, it watches only your protected locations, it does not stop processes or block file writes, and it cannot decrypt or recover files. It cannot guarantee that ransomware is prevented. These limits are described in section 16.

## The workflow

| Stage              | Where                                           | What you do                                                         |
|--------------------|-------------------------------------------------|---------------------------------------------------------------------|
| <b>Protect</b>     | Dashboard · Protection · Protected Locations    | Choose folders, start protection, keep it running.                  |
| <b>Investigate</b> | Activity · Alerts · Incidents                   | See what changed, why it was flagged, and which case it belongs to. |
| <b>Respond</b>     | Incident detail · Quarantine · Recovery · Trust | Acknowledge, investigate, contain, resolve; check your backups.     |
| <b>Verify</b>      | Diagnostics · Logs                              | Check component health and the audit chain.                         |

Each page shows a one-line hint under its title with its stage (Protect › Investigate › Respond › Verify), what to do there, and a link to the natural next page.

## 2. First launch

Start Sentinel with `run.ps1` (see `SETUP.md`) or with `python main.py` from the source folder. On the first launch the main window opens with the setup wizard over it.

### Step 1 — Welcome

The first screen says in four lines what Sentinel does and, just as plainly, what it does not do.

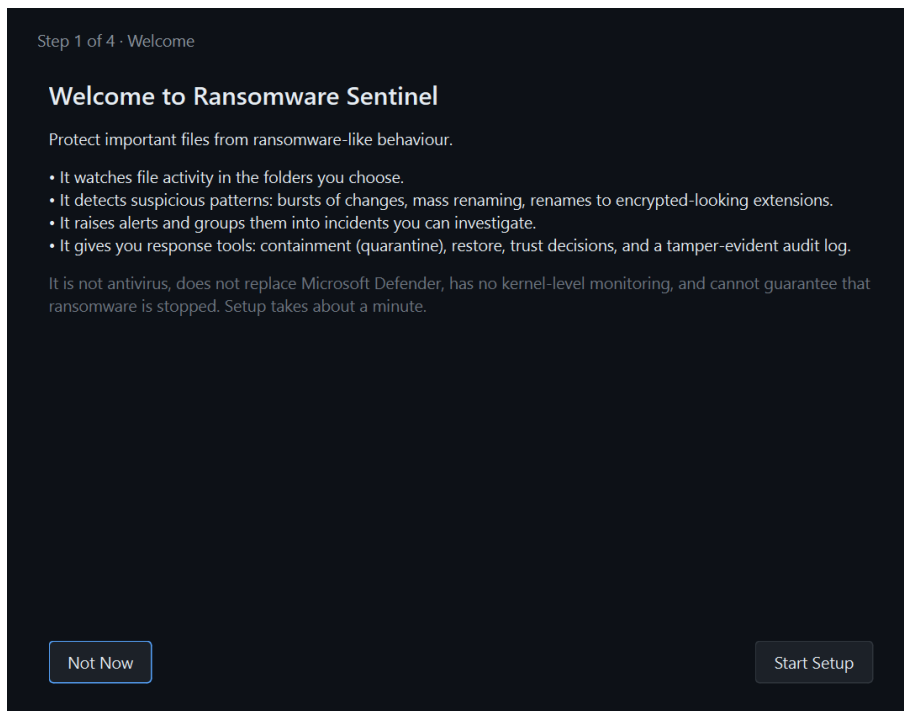


Figure 1. Setup, step 1: Welcome

1. Read the summary.
2. Press Start Setup. (Not Now closes setup; reopen it later from Help → Run Setup Again.)

## Step 2 — Protected locations

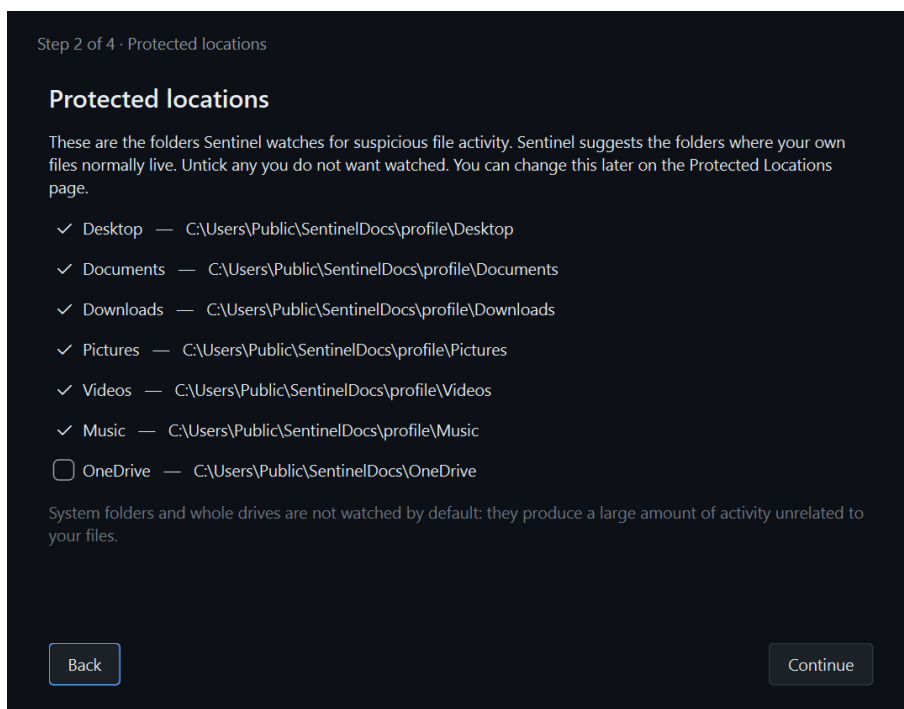


Figure 2. Setup, step 2: the proposed folders

Sentinel lists the folders where your own files normally live, ticked: Desktop, Documents, Downloads, Pictures, Videos and Music, found through Windows itself, so a Documents folder moved to OneDrive or another drive is found where it really is. If OneDrive is set up it is listed unticked, because its sync client creates and renames files in bursts that can raise alerts.

3. Untick any folder you do not want watched.
4. Press Continue.

### Step 3 — Initial scan

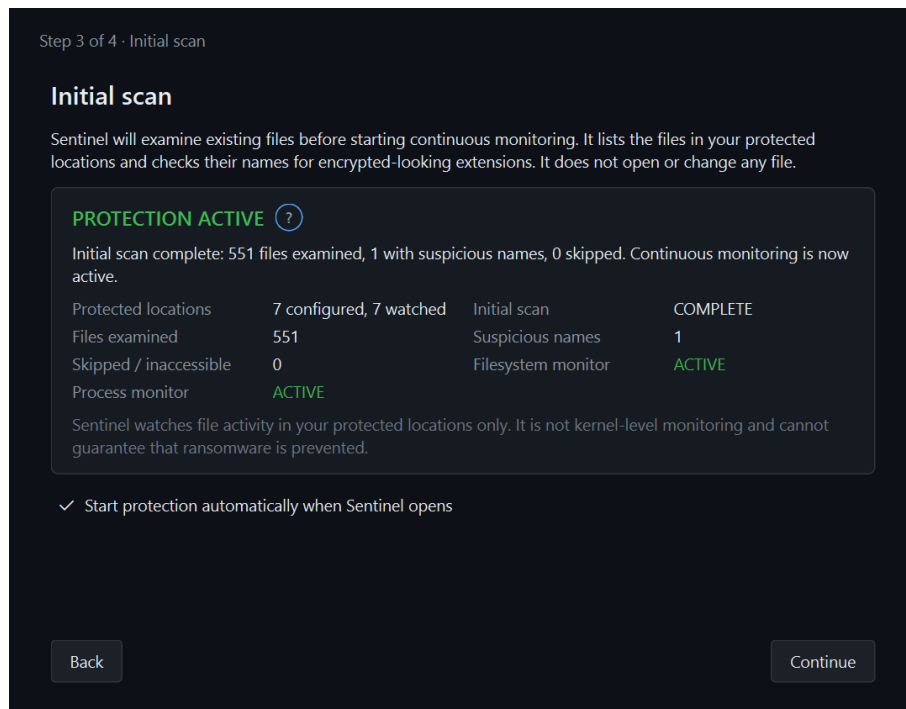


Figure 3. Setup, step 3: initial scan complete, protection active

5. Leave “Start protection automatically when Sentinel opens” ticked if you want protection to start by itself next time.
6. Press Start Protection. The initial scan runs; Skip Scan stops it early (monitoring still starts).
7. When the panel says PROTECTION ACTIVE, press Continue.

### Step 4 — Protection active

The last screen shows what is running and offers three next steps: Take Guided Tour, Explore Dashboard, or Run Safe Demo. The first time you finish setup, the guided tour starts by itself unless you choose the demo.

## The guided tour

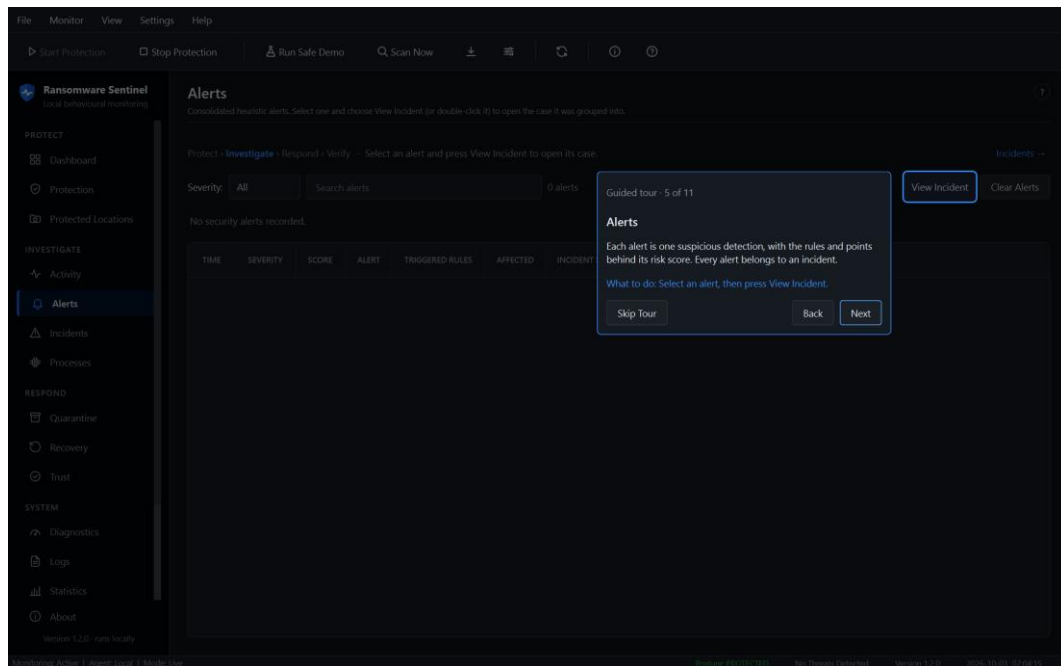


Figure 4. The guided tour points at the real control on each page

The tour dims the window, outlines one real control and explains what it is and what to do with it. Use Next, Back and Skip Tour, the arrow keys, or Esc. It ends on the Dashboard. It starts by itself only once; replay it any time from Help → Guided Tour or Getting Started.

## 3. Protected Locations

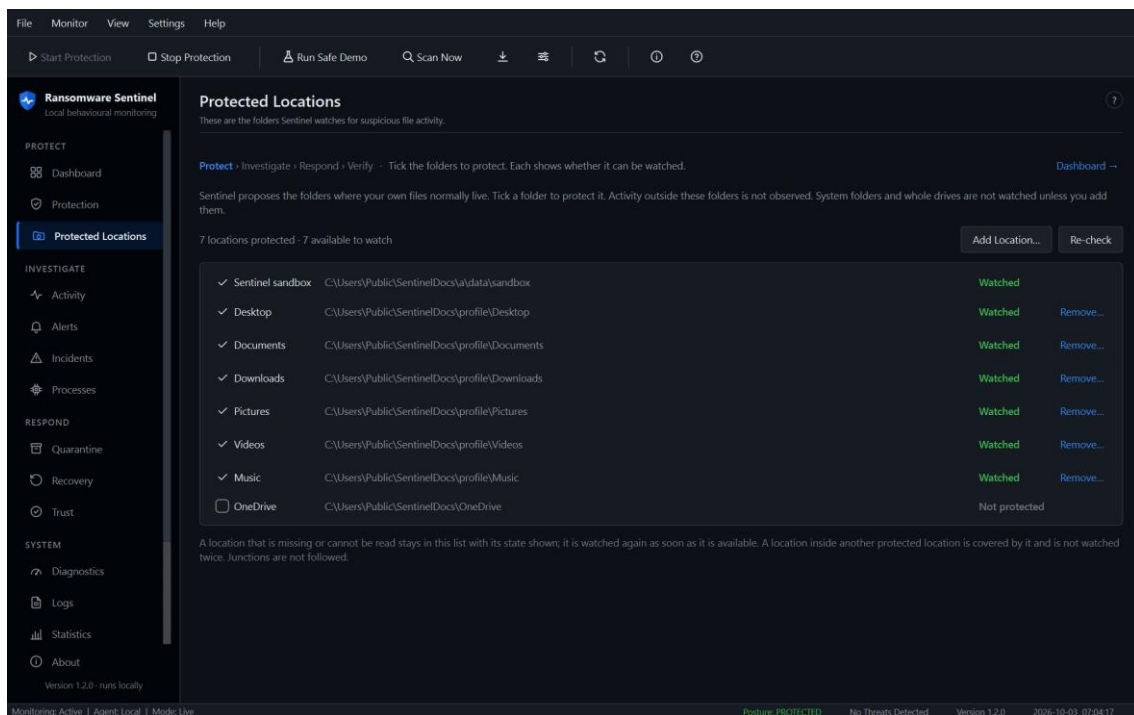


Figure 5. Protected Locations

This page lists every protected folder and whether it can actually be watched.

| State                                       | Meaning                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Watched</b>                              | Activity in this folder is observed while protection runs.                                                              |
| <b>Covered</b>                              | This folder is inside another protected folder, which is watched instead (it is not watched twice).                     |
| <b>Missing</b>                              | The folder does not exist now (for example a disconnected drive). It stays listed and is watched again when it returns. |
| <b>No access</b>                            | Sentinel does not have permission to read it.                                                                           |
| <b>Junction – not followed</b>              | The folder is a link that could point elsewhere; it is not followed.                                                    |
| <b>Off</b>                                  | You unticked it. It stays in the list.                                                                                  |
| <b>Not protected / Not on this computer</b> | A suggested folder you have not added, or one that does not exist.                                                      |

- **Tick or untick** a folder to protect it or stop watching it. Unticking keeps it in the list.
- **Add Location...** adds any other folder. Adding a whole drive or a Windows/Program Files folder shows a warning first, because it produces far more activity and permission errors.
- **Remove...** takes a folder off the list. The Sentinel sandbox is always watched and cannot be removed.
- **Re-check** asks every folder for a fresh answer.

## 4. Initial scan

Whenever protection starts, Sentinel first lists the files already in your protected locations and checks their names for encrypted-looking extensions. It does not open, read or change any file. The scan runs before continuous monitoring starts, so listing files is never mistaken for ransomware activity.

| Shown                         | Meaning                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Files examined</b>         | How many files were listed.                                                                               |
| <b>Suspicious names</b>       | Files whose extension is on the suspicious list (for example .locked). A name is not proof of encryption. |
| <b>Skipped / inaccessible</b> | Folders that could not be read and were passed over.                                                      |
| <b>Initial scan</b>           | NOT RUN, RUNNING, COMPLETE, CANCELLED or FAILED.                                                          |

Skip Scan stops the scan early; continuous monitoring still starts. Junctions are not followed.

## 5. Protection active

Protection moves through clear states, shown on the Dashboard and in setup:

| State | Meaning |
|-------|---------|
|-------|---------|

|                               |                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------|
| <b>PROTECTION OFF</b>         | Nothing is being watched. Press Start Protection.                                          |
| <b>STARTING PROTECTION</b>    | Protected locations are being checked (at most about 15 seconds for slow network folders). |
| <b>INITIAL SCAN</b>           | Existing files are being listed.                                                           |
| <b>PROTECTION ACTIVE</b>      | Continuous monitoring of your protected locations is running.                              |
| <b>STOPPING</b>               | Protection is being stopped.                                                               |
| <b>PROTECTION NOT STARTED</b> | Something failed; the message says what (for example no folder can be watched).            |

Start and stop protection with the toolbar (Ctrl+Shift+M / Ctrl+Shift+S) or the Dashboard panel. "Protection active" means monitoring of your protected locations is running. It is not kernel-level monitoring and not a guarantee.

## 6. Dashboard

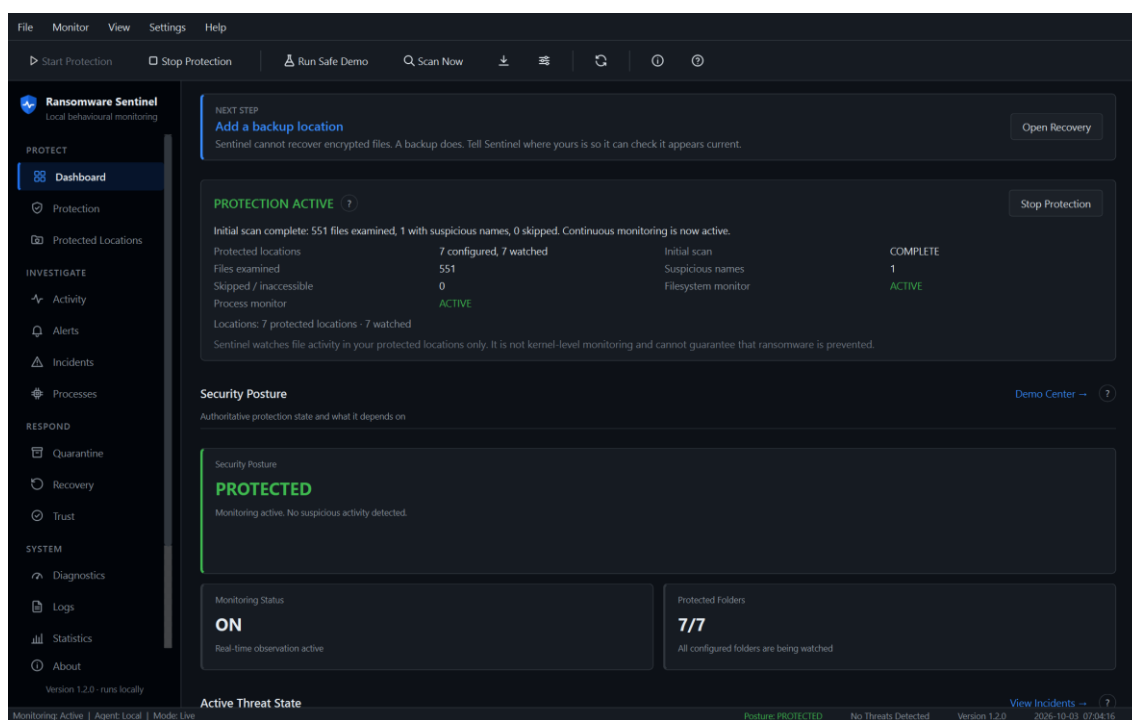


Figure 6. Dashboard with protection active

- **Next step.** One recommended action, decided from the current state: set up protection, start it, open an incident, fix a location, add a backup location, or try the demo. Press its button to do it.
- **Protection panel.** State, protected locations (watched, covered, missing...), the initial scan results, and whether the filesystem and process monitors are active.
- **Security Posture.** Sentinel's overall verdict on its own monitoring: PROTECTED, MONITORING, DEGRADED, ERROR, CRITICAL THREAT or MONITORING OFF. PROTECTED means the pipeline is working on your protected locations, not that the computer is safe.
- **Active Threat State.** The current risk score and the number of active and critical incidents.
- **Recent alerts and activity.** The latest alerts and file changes.

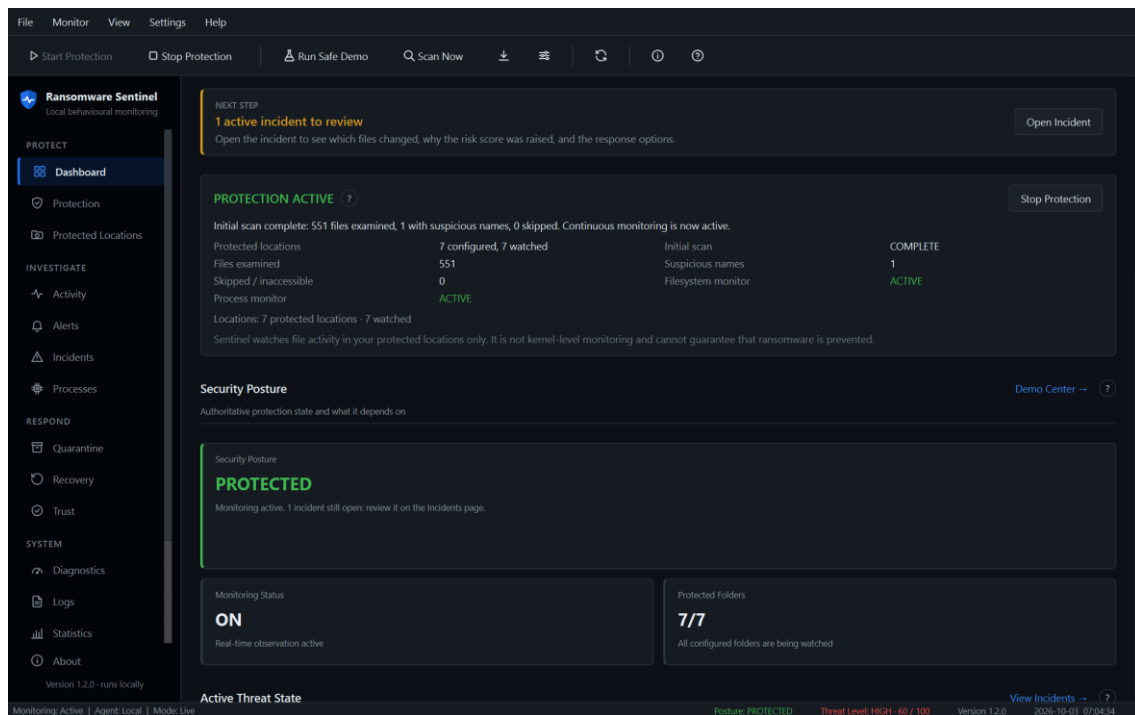


Figure 7. Dashboard when an incident needs attention

## 7. Alerts

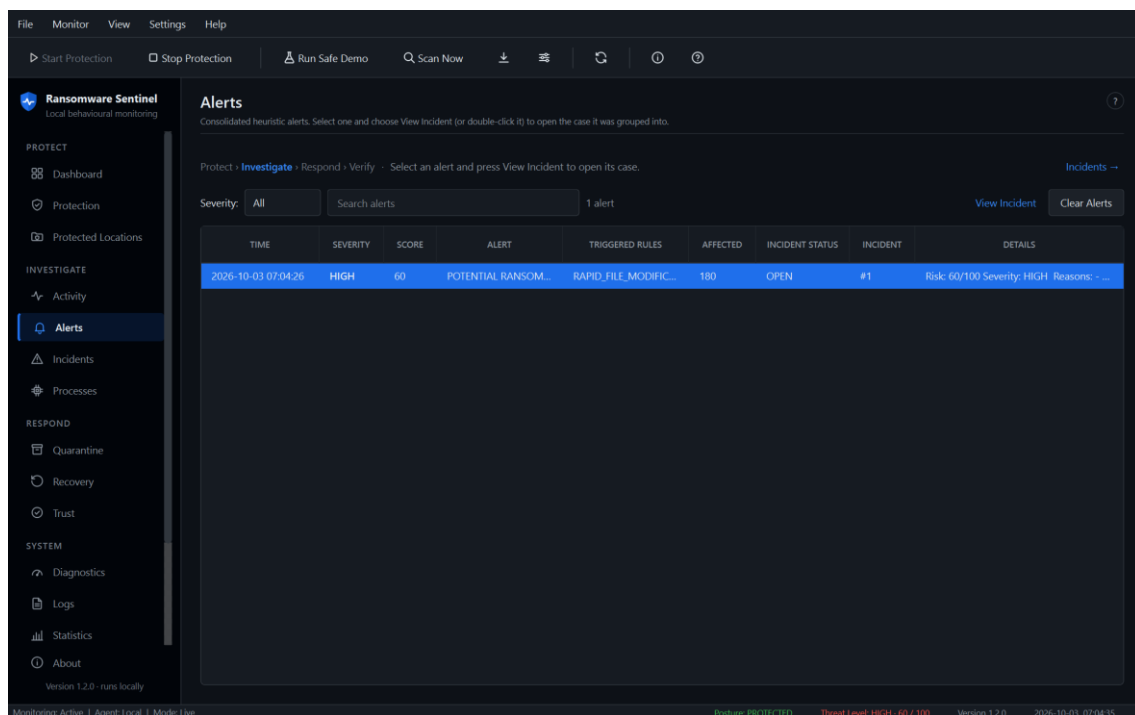


Figure 8. Alerts

An alert is one suspicious detection. Each row shows its severity, risk score, the rules that fired, how many files were affected, and the incident it belongs to.

8. Select an alert.
9. Press View Incident (or double-click the row) to open its case.

## 8. Incidents

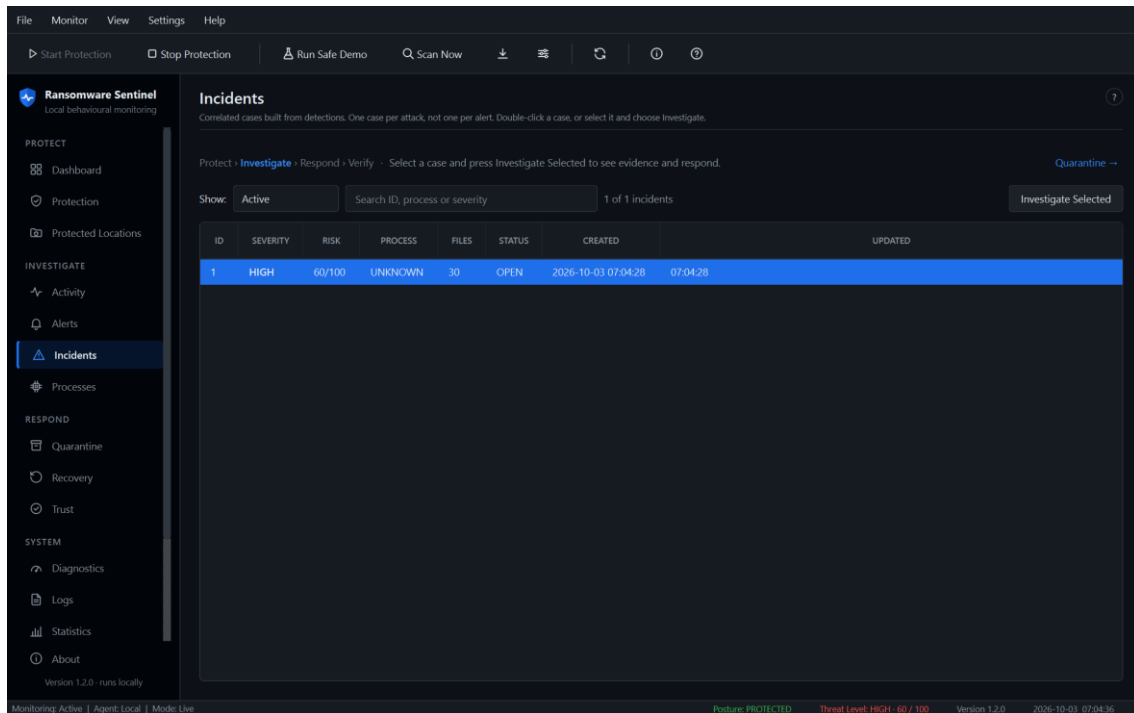


Figure 9. Incidents

An incident groups the alerts from one episode into one case. Active cases are listed first; use Show to see resolved and dismissed ones too.

10. Select a case.

11. Press Investigate Selected (or double-click it).

| Status                         | Meaning                                               |
|--------------------------------|-------------------------------------------------------|
| <b>OPEN</b>                    | New; nobody has acted on it yet.                      |
| <b>INVESTIGATING</b>           | You marked it as being looked into.                   |
| <b>CONTAINED / QUARANTINED</b> | A file attributed to it was contained.                |
| <b>RESOLVED</b>                | Closed after investigation.                           |
| <b>DISMISSED</b>               | Closed as not a threat (for example a demo incident). |

## 9. Investigation

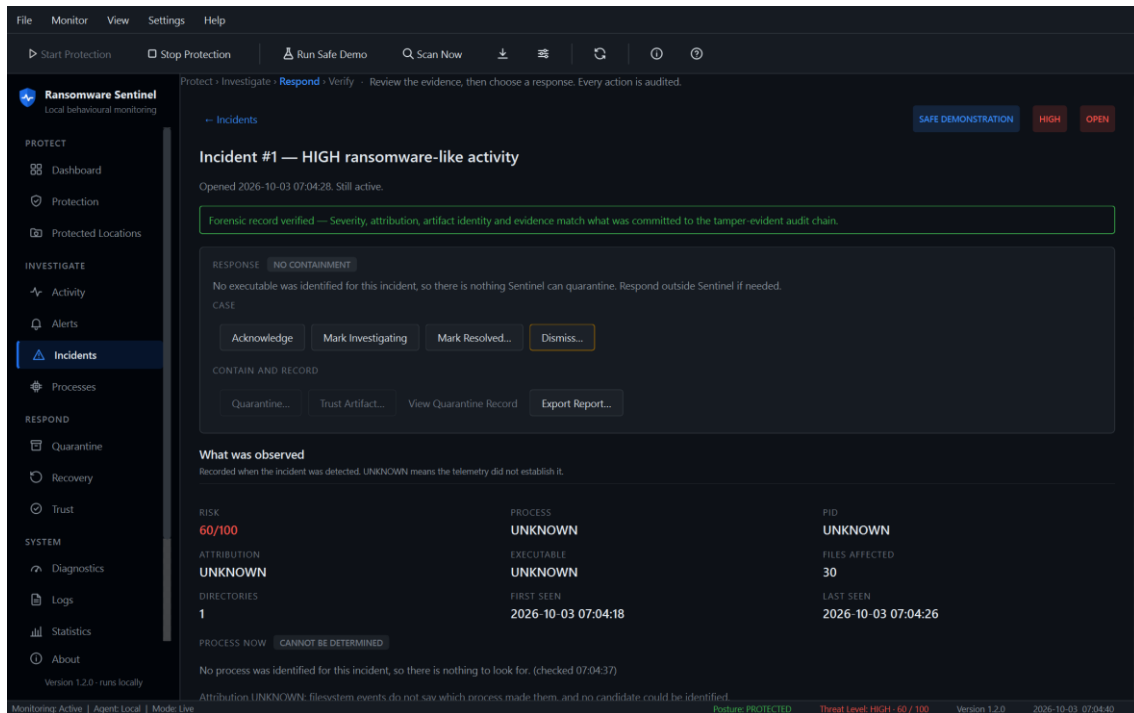


Figure 10. Incident detail (a Demo Center incident)

- **Forensic record.** A banner says whether the incident's stored record still matches what was committed to the audit chain.
- **Response.** Where containment stands and every available action.
- **What was observed.** Risk, process and attribution, files affected, first and last seen. Attribution is usually UNKNOWN: file events do not say which process made a change, and Sentinel does not guess.
- **Why this is suspicious.** The rules and points behind the score.
- **File activity.** What happened to each file, following renames as one file.
- **Timeline.** Every recorded action, with who did it and the real outcome.

Actions: Acknowledge (you have seen it), Mark Investigating, Mark Resolved..., Dismiss..., Quarantine..., Trust Artifact..., Export Report.... Every action is confirmed and recorded in the incident's audit trail.

## 10. Quarantine

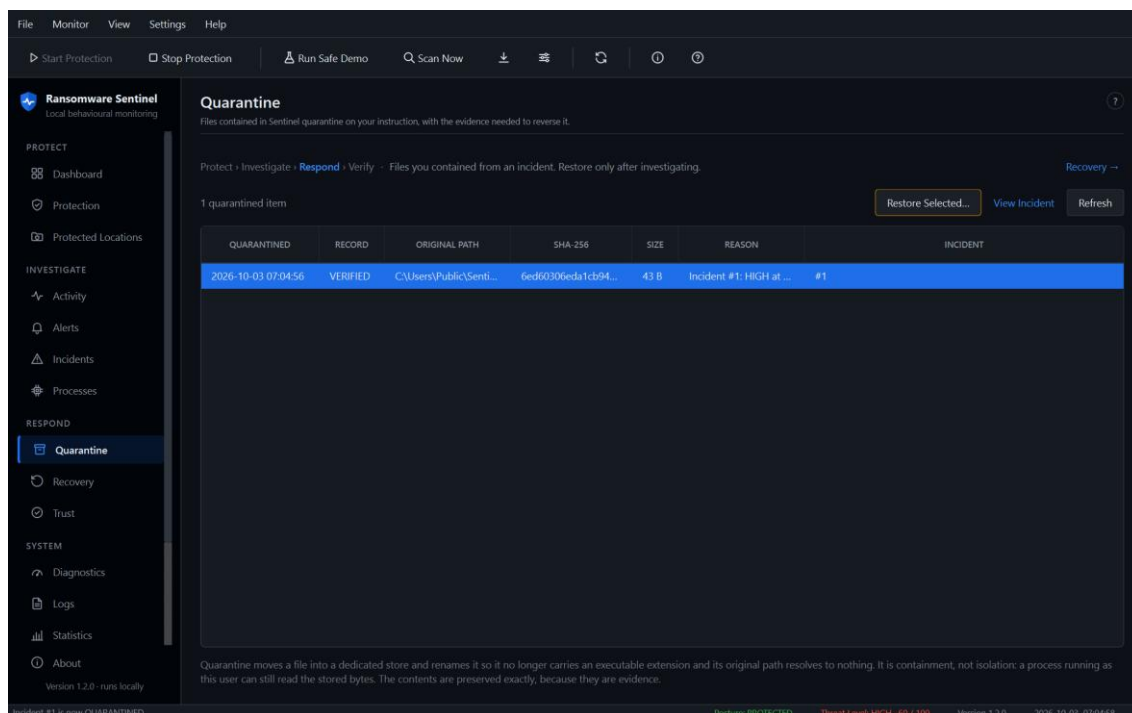


Figure 11. Quarantine (documentation data: a harmless file stands in for an attributed executable)

Quarantine is available when an incident names an executable. It moves that file into Sentinel's quarantine store and renames it, so it no longer has an executable extension and its original path no longer resolves.

**Containment, not isolation.** A program running as you can still read the stored bytes, and anything with higher privileges is not contained. Sentinel does not stop the process.

12. To restore, select the item and press Restore Selected.... Sentinel checks the file is byte-for-byte the one it quarantined and never overwrites an existing file.
13. View Incident opens the case the item came from.

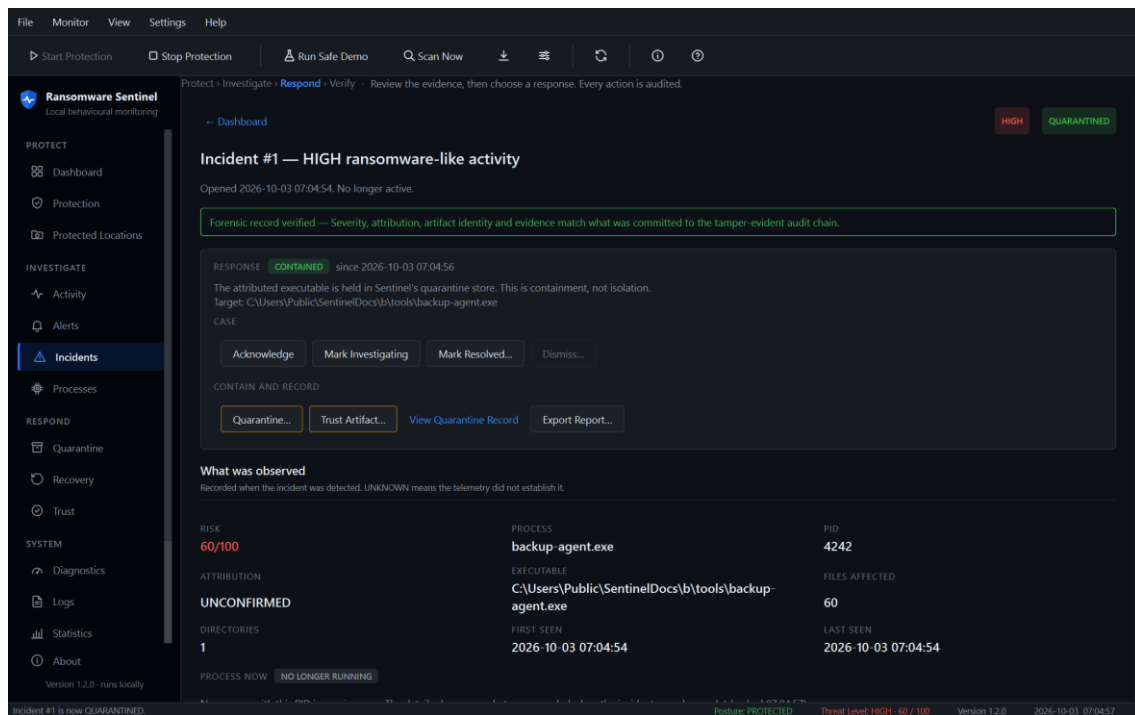


Figure 12. An incident after quarantine (documentation data)

## 11. Recovery readiness

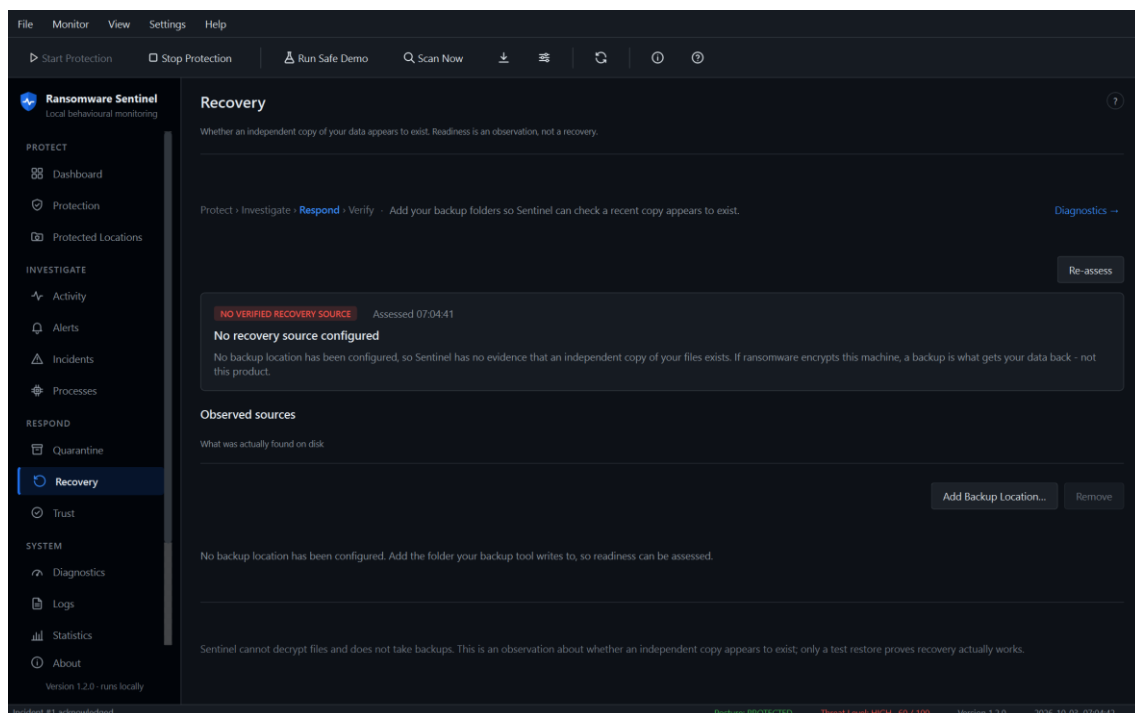


Figure 13. Recovery

Sentinel cannot recover encrypted files; a backup can. This page checks whether a recent, independent copy of your data appears to exist in the backup locations you add.

14. Press Add Backup Location... and choose the folder your backup tool writes to.
15. Press Re-assess. The result is RECOVERY READY, RECOVERY PARTIAL, NO VERIFIED RECOVERY SOURCE or RECOVERY UNKNOWN.

This is an observation, not a recovery. Only a test restore proves that a backup works.

## 12. Diagnostics and audit

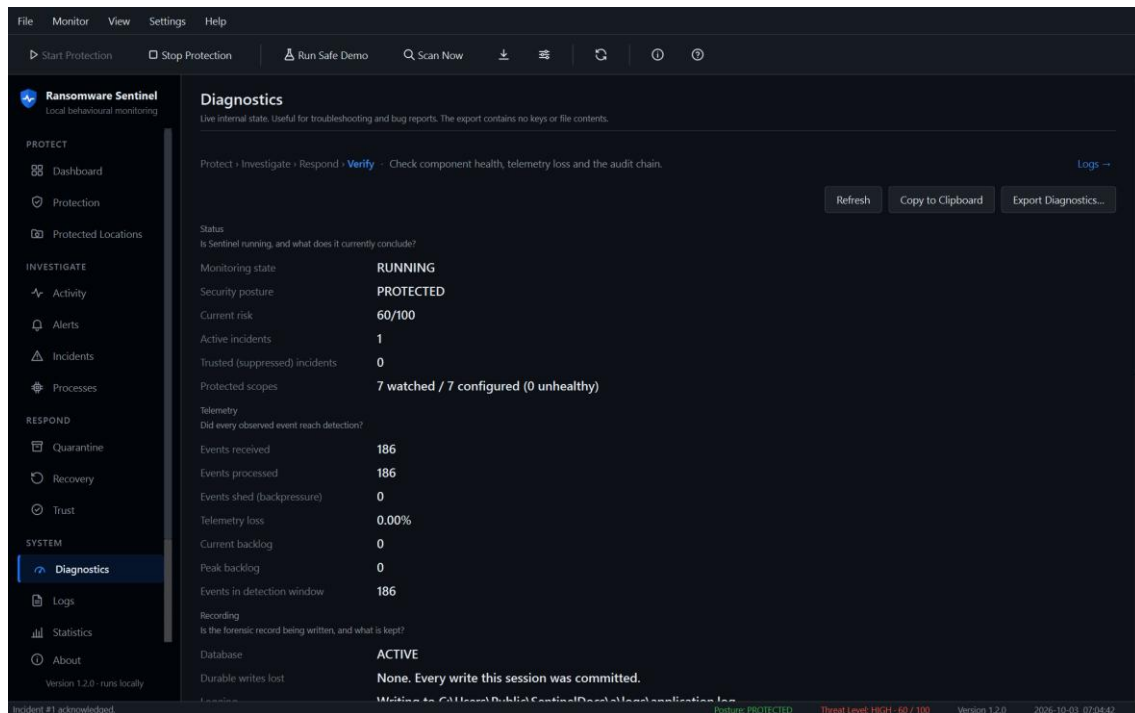


Figure 14. Diagnostics

Diagnostics shows whether Sentinel itself is working: monitoring state, posture, events received, processed and shed, database state, retention, and the integrity of the audit chain and configuration.

- **Verify Now** re-checks the HMAC-chained audit log. A failure means stored security records may have been changed outside Sentinel.
- **Export Diagnostics... / Copy to Clipboard** for support. The export contains no keys or file contents; it does contain file paths.
- **Logs** lists every security action and event with its level and source; File → Export Logs saves the log files.

## 13. Safe Demo (Demo Center)

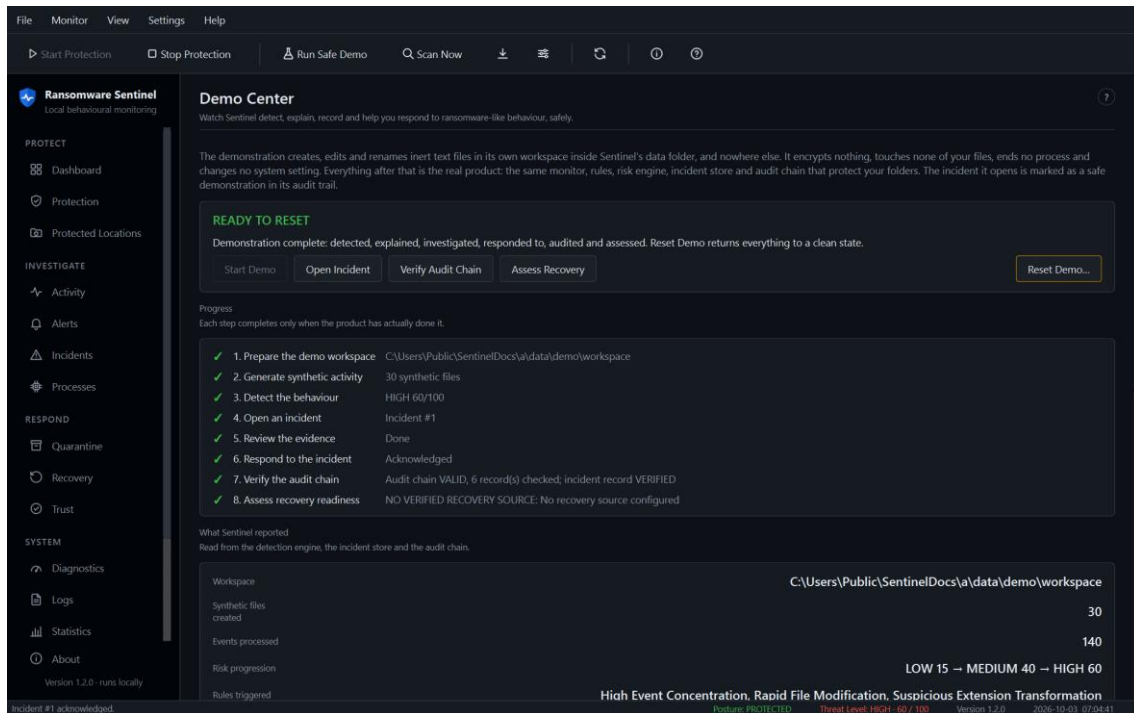


Figure 15. Demo Center after a complete demonstration

The Demo Center is the safe way to see, or to show someone, the whole workflow. It creates, edits and renames inert text files in its own isolated workspace inside Sentinel's data folder, and nowhere else. It encrypts nothing, touches none of your files, ends no process and changes no system setting. Everything after that is the real product.

16. Open Demo Center (Learn group in the sidebar).
17. If an earlier run left anything behind, press Reset Demo... first.
18. Press Start Demo. If protection is off, the demo starts monitoring and stops it again on reset.
19. Follow the Progress list; each step turns green only when the product has actually done it.
20. Use the Presenting it buttons to show the Activity, Dashboard, Alerts and the incident.
21. Open the incident and press Acknowledge, then return and press Verify Audit Chain and Assess Recovery.
22. Press Reset Demo... and confirm. The synthetic files are removed and the demo incident is dismissed; records are kept.

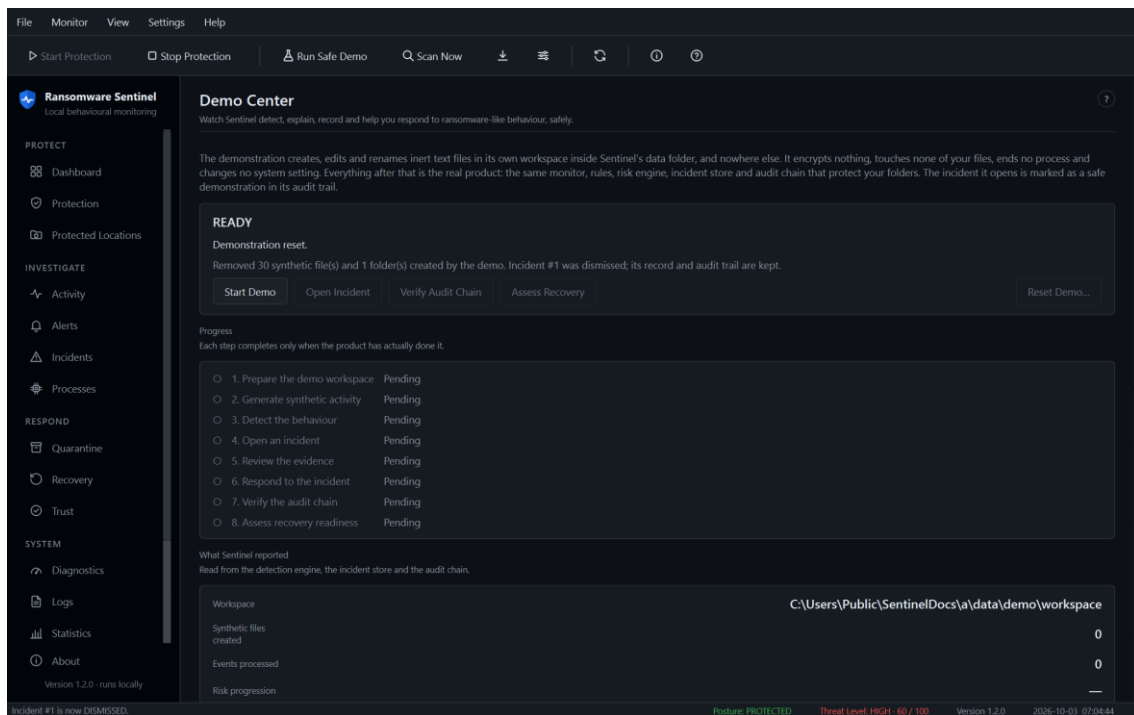


Figure 16. After reset: the workspace is clean

Start Demo is refused while another incident is active, so a demo alert can never be merged into a real case. The demo incident has no process to attribute, so quarantine and trust are not available for it — that is the product's real attribution limit, not a demo shortcut.

## 14. Settings

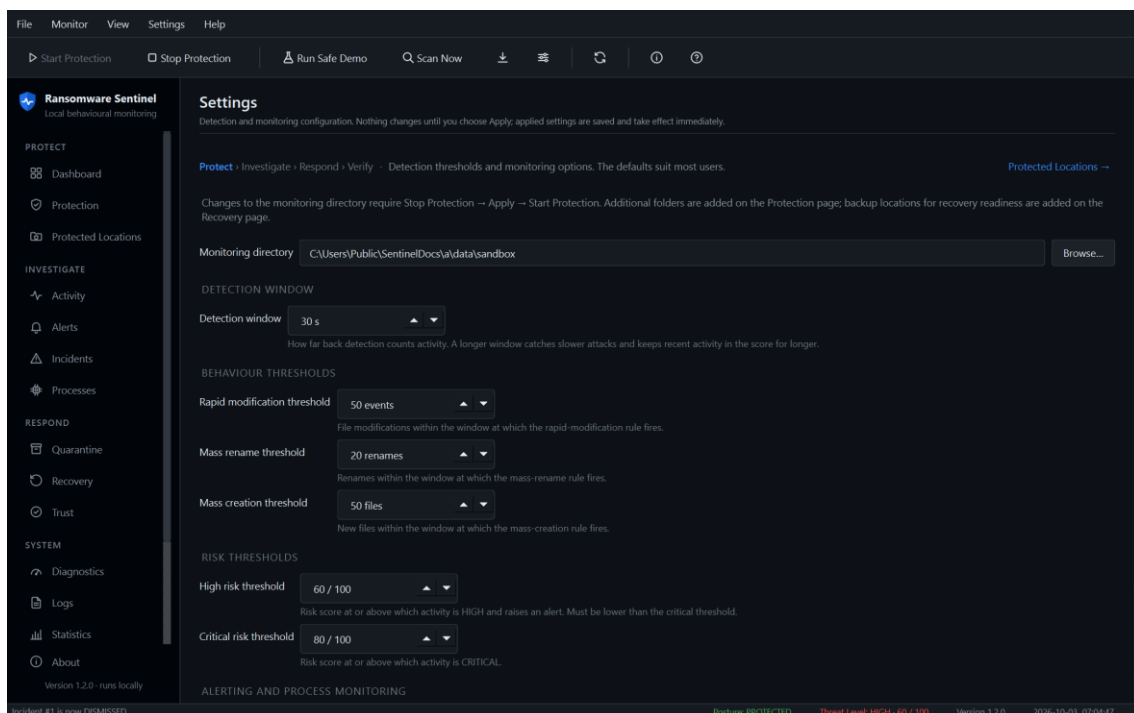


Figure 17. Settings

Settings holds the detection window, the behaviour thresholds, the risk thresholds and the alert cooldown. The defaults suit most users. Nothing changes until you press Apply; applied settings are

saved and take effect immediately. Protected folders are managed on the Protected Locations page, and backup folders on the Recovery page.

## 15. Troubleshooting

| Problem                                         | What to do                                                                                                                             |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>"Ransomware Sentinel is already running"</b> | Only one copy may use a data folder. Close the other window (check the taskbar).                                                       |
| <b>A protected location shows Missing</b>       | Reconnect the drive or recreate the folder; press Re-check. It stays listed.                                                           |
| <b>A location shows No access</b>               | Sentinel runs with your permissions; choose a folder you can open.                                                                     |
| <b>Protection did not start</b>                 | Read the message on the Dashboard panel, then open Protection to see which component failed.                                           |
| <b>Posture DEGRADED</b>                         | Open Protection or Diagnostics; it names what is not fully working (for example a folder still being checked or telemetry being shed). |
| <b>Start Demo is unavailable</b>                | Resolve or dismiss active incidents first, or press Reset Demo... to clear an earlier run.                                             |
| <b>Many alerts from OneDrive</b>                | Untick OneDrive on Protected Locations; its sync activity resembles bursts of renames.                                                 |
| <b>Audit verification failed</b>                | Records may have been changed outside Sentinel. Export Diagnostics for support.                                                        |
| <b>The setup or tour was dismissed</b>          | Help → Run Setup Again or Help → Guided Tour; Help → Quick Reference (F1).                                                             |

## 16. Limitations

- Not antivirus and not a replacement for Microsoft Defender or an EDR product.
- User-mode only: no kernel driver, ETW consumer or minifilter.
- Only protected locations are watched; activity anywhere else is not observed.
- Process attribution is usually UNKNOWN; a named process is a candidate, not proven cause.
- No process termination or write blocking: Sentinel detects and records; you act.
- No decryption and no recovery of encrypted files; recovery readiness only checks for a backup.
- Quarantine is containment, not isolation.
- The audit chain detects tampering afterwards; it cannot prevent it.
- During the initial scan, monitoring is not yet running.
- It cannot guarantee that every ransomware family is detected.